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Chapter 5

Servers and Virtualization

Servers are the lifeblood of any network. They provide the shared resources that network users crave, such as file storage, databases, email, web services, and so on. Choosing which servers your network needs and selecting the type of equipment you use to implement your servers are among the key decisions you'll make when you set up a network.

In this chapter, I take a quick look at what's important in a server. First, I cover the basic functions of a server operating system. Then I survey the various types of servers most networks need. Then I turn my attention to important matters to consider when selecting the kind of hardware a server should run on. And finally, I look quickly at the idea of virtualizing your entire server environment.

Understanding Network Operating Systems

The server operating system is what enables your server computers to function as servers rather than as ordinary Windows clients. Server operating systems provide essential functions such as providing basic security services, sharing disk

storage and printers, and so on. The following sections cover some of the core functions of a server operating system.

Network services

Obviously, a server operating system must provide networking capabilities in order for it to function on a network. If your client computers can't connect to your servers, your network will be useless. For this reason, it's a good idea to make sure your server computers are equipped with more than one network interface. That way, if one of the interfaces fails, the other can pick up the slack and keep your server connected to your network.

In addition to basic network connectivity, one of your servers will typically be responsible for providing some essential software services that are required to keep a network operating in an efficient manner. One of these is called Dynamic Host Configuration Protocol (DHCP); it's the service that recognizes computers and other devices that want to join the network, providing each with a unique address so that all the devices on the network can identify one another. For more information about this vital service, refer to Book 2, Chapter 5.

A second basic service that is provided by one of the servers on your network is called Domain Name System (DNS). This service is what enables people to use network names instead of the actual addresses that are handed out by DHCP. It's also the service that enables people to browse the World Wide Web using addresses such as `www.amazon.com` rather than cryptic addresses like `54.239.28.85`. For more information about this important service, refer to Book 2, Chapter 6.

File-sharing services

One of the most important functions of a server operating system is to share resources with other network users. The most common resource that's shared is the server's *file system* (organized disk space that a server must be able to share, in whole or in part, with other users). In effect, users can treat the server's disk space as an extension of their own computers' disk space.

The server operating system allows the system administrator to determine which portions of the server's file system to share.



TIP

Although an entire hard drive can be shared, it isn't commonly done. Instead, individual folders are shared. The administrator can control which users are allowed to access each shared folder.

Because file sharing is the reason many network servers exist, server operating systems have more sophisticated disk management features than are found in desktop operating systems. For example, most server operating systems can manage two or more hard drives as though they were a single drive. In addition, most can create a *mirror* (an automatic backup copy of a drive) on a second drive.

Windows server operating systems also provide a feature called *Distributed File System* (DFS). DFS lets you spread your file shares across multiple servers while creating a naming scheme that lets you access the data without referencing a specific server. This makes it much easier to reorganize your file shares to accommodate growing file storage needs. For more about DFS, refer to Book 6, Chapter 5.

Multitasking

Only one user at a time uses a desktop computer; however, multiple users simultaneously use server computers. As a result, a server operating system must provide support for multiple users who access the server remotely via the network.

At the heart of multiuser support is *multitasking*, which is the capability of an operating system to execute more than one program (a task or a process) at a time. Multitasking operating systems are like the guy who used to spin plates balanced on sticks on the old *Ed Sullivan Show* back in the 1950s. He'd run from plate to plate, trying to keep them all spinning so they wouldn't fall off the sticks — and just for grins, he was blindfolded or riding on a unicycle.

Although multitasking creates the appearance that two or more programs are executing on the computer at one time, in reality, a computer with a single processor can execute only one program at a time. The operating system switches the central processing unit (CPU) from one program to another to create the appearance that several programs are executing simultaneously, but at any given moment, only one of the programs is actually executing; the others are patiently waiting for their turns. (However, if the computer has more than one CPU, the CPUs *can* execute programs simultaneously, which is called *multiprocessing*.)

For multitasking to work reliably, the server operating system must completely isolate the executing programs from each other. Otherwise, one program may perform an operation that adversely affects another program. Multitasking operating systems do this by providing each task with its own unique address space that makes it almost impossible for one task to affect memory that belongs to another task.

Directory services

Directories are everywhere — and they were even in the days when they were all hard copies. When you needed to make a phone call, you looked up the number in a phone directory. When you needed to find the address of a client, you looked them up in your Rolodex. And then there were the nonbook versions: When you needed to find the Sam Goody store at a shopping mall, you looked for the mall directory — usually, a lighted sign showing what was where.

Networks have directories, too, providing information about the resources that are available on the network: users, computers, printers, shared folders, and files. Directories are essential parts of any server operating system.

The most popular modern directory service is called *Active Directory*. Active Directory is a standard component of all Windows operating systems, and because it's so popular, most other operating systems support it as well. Active Directory is a database that organizes information about a network and all its computers and users. It's simple enough to use for networks with just a few computers and users, but powerful enough to work with large networks containing tens of thousands of computers and users. Figure 5-1 shows the Active Directory Users and Computers tool, which manages Active Directory user and computer accounts on Windows Server 2025.

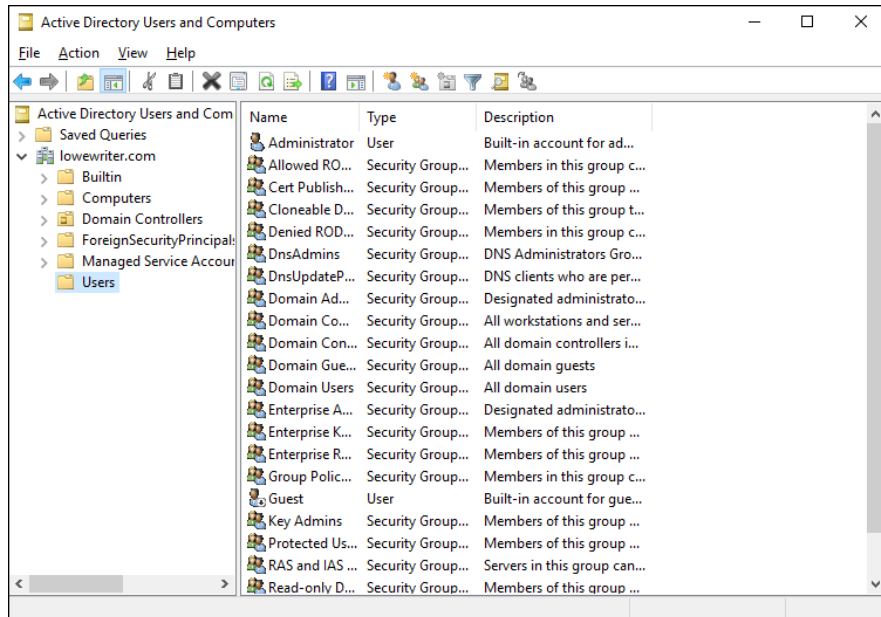


FIGURE 5-1:
Managing Active
Directory users
and computers.

Security services

All server operating systems must provide some measure of security to protect the network from unauthorized access. Hacking seems to be the national pastime these days. With most computer networks connected to the internet, anyone anywhere in the world can — and probably will — try to break into your network.

The most basic type of security is handled through *user accounts*, which grant individual users the right to access the network resources and govern which resources the user can access. User accounts are secured by passwords; therefore, good password policy is a cornerstone of any security system. Most server operating systems give you some standard tools for maintaining network security:

- » **Establish password policies.** For example, you can mandate that passwords have a minimum length and include a mix of letters and numerals.
- » **Set passwords to expire after a certain number of days.** Network users must change their passwords frequently.
- » **Encrypt network data.** A data-encryption capability scrambles data before it's sent over the network or saved on disk, making unauthorized use a lot more difficult.



TIP

Good encryption is the key to setting up a virtual private network (VPN), which enables network users to securely access a network from a remote location by using an internet connection.

- » **Manage digital certificates.** Digital certificates are used to ensure that users are who they say they are and files are what they claim to be.

The overwhelming majority of business networks rely on server versions of Windows, known as Windows Server. Microsoft periodically releases updated versions of Windows Server, so Windows Server is frequently improved, and older versions are occasionally rendered obsolete. Currently, the most commonly used versions are Windows Server 2019 and Windows Server 2012. The newest version is known as Windows Server 2025; you'll learn about Windows Server 2025 throughout this entire book, but Book 6 dives deeply into setting it up and configuring it.

Windows Server is not the only server operating system at your disposal. Many servers — especially those whose primary responsibility is to host websites — use Linux instead of Windows Server. You can find out more about Linux in Book 8.

Knowing What's Important in a Server

The following sections point out some general things to keep in mind when selecting the equipment that a server should run on.

Scalability

Scalability is the ability to increase the size and capacity of the server computer without unreasonable hassle. Purchasing a server computer that just meets your current needs is a major mistake because, rest assured, your needs will double within a year. If at all possible, equip your servers with far more disk space, random access memory (RAM), and processor power than you currently need.

Reliability

The old adage “You get what you pay for” applies especially well to server computers. Why spend \$5,000 on a server computer when you can buy one with seemingly similar specifications at a discount electronics store for a mere \$1,000? The main reason: reliability. When a client computer fails, only the person who uses that computer is affected. When a server fails, however, everyone on the network is affected. The less-expensive computer is probably made of inferior components that are more likely to fail, and it doesn't have redundant components built in. For example, many server computers have two power supplies, two CPUs, two or more network interfaces, and other redundant components.

Availability

Availability is closely related to reliability. When a server computer fails, how long does it take to correct the problem and get the server up and running again? Server computers are designed so their components can be easily diagnosed and replaced, which minimizes the downtime that results when a component fails. In some servers, components are *hot swappable* (certain components can be replaced without shutting down the server). Some servers are fault tolerant so that they can continue to operate even if a major component fails.

Service and support

Service and support are often overlooked factors when picking computers. If a component in a server computer fails, do you have someone on-site qualified to repair the broken computer? If not, you should get an on-site maintenance contract for the computer.



WARNING

Don't settle for a maintenance contract that requires you to take the computer in to a repair shop or, worse, mail it to a repair facility. You can't afford to be without your server that long. Get a maintenance contract that provides for on-site service and repair of your server 24 hours a day, seven days a week.

Identifying the Components of a Server Computer

The hardware components that make up a typical server computer are similar to the components used in less-expensive client computers. However, server computers are usually built from higher-grade components than client computers for the reasons given in the “Knowing What’s Important in a Server” section, earlier in this chapter. The following sections describe the typical components of a server computer.

Motherboard

A motherboard is the computer's main electronic circuit board to which all the other components of your computer are connected. More than any other component, the motherboard *is* the computer. All other components attach to the motherboard.

The major components on the motherboard include the CPU, supporting circuitry (the chipset), RAM, expansion slots, a hard drive controller, USB ports for devices such as keyboards and mice, a graphics adapter, and one or more network interfaces.

Processor

The CPU is the brain of the computer. Although the processor isn't the only component that affects overall system performance, it's the one that most people think of first when deciding what type of server to purchase. At the time of this writing, most servers used one of several variations of Intel's Xeon processor. These processors are designed specifically for server computers rather than client computers; they offer anywhere from 4 to 22 independent processor cores, depending on the model.

Each motherboard is designed to support a particular type of processor. CPUs come in two basic mounting styles: slot or socket. However, you can choose from several types of slots and sockets, so you have to make sure that the motherboard supports the specific slot or socket style used by the CPU. Some server motherboards have two or more slots or sockets to hold two or more CPUs.



Clock speed refers to how fast the basic clock that drives the processor's operation ticks. In theory, the faster the clock speed, the faster the processor. However, clock speed alone is reliable only for comparing processors within the same family. What matters more in a server is the number of processor cores. The more cores the server has, the more tasks the server can perform simultaneously. Because servers are in the business of supporting many clients, being able to do many tasks simultaneously is a huge benefit for server performance.

What's more, processor cores utilize a technology called *hyperthreading*, which effectively lets each processor core juggle two threads at once. (In general terms, a *thread* is a sequence of instructions that performs a single task.) Because each core can handle two simultaneous threads, a processor with four cores can handle eight concurrent threads.

Many server motherboards can support two separate processors, which doubles the potential workload of the server. For example, if the server has two 14-core processors, the server has a total of 28 cores available for its workload. Because of hyperthreading, each of these 28 cores can handle 2 threads, so the server can handle 56 concurrent threads.

Memory

Don't scrimp on memory. People rarely complain about servers having too much memory. The total memory capacity of the server depends on the motherboard. It isn't unusual to see servers configured with anywhere from 32GB to 512GB of RAM.

Hard drives

Most desktop computers use inexpensive consumer-grade SATA hard drives, which are adequate for individual users. Because of their low cost, SATA drives are sometimes also used in inexpensive servers. But because performance and reliability are important in servers, most servers rely on faster and more reliable SCSI or SAS disk drives instead. For the best performance, solid-state drives (SSDs) can be used; these drives have no mechanical parts, so they're considerably faster than traditional spinning disks.

Network interfaces

The network connection is one of the most important parts of any server. Ideally, your server should have at least two network interfaces. Additional network interfaces not only improve the performance of your server, but also make it more reliable. If one of the network interfaces fails, the other(s) can pick up the ball.

If possible, the server's network interfaces should be 10 gigabit per second (Gbps) interfaces. Then, you can use 10 Gbps switches to connect the servers to each other and to your access switches. With many users contending for access to the servers simultaneously, 1 Gbps interfaces can easily become a performance-limiting bottleneck.

Video

Fancy graphics aren't that important for a server computer. You don't need to equip your server with an expensive video card; the video interface that's built in to the motherboard will suffice. (This is one of the few areas where cutting costs on a server is okay.)

Power supply

Because a server usually has more devices than a typical desktop computer, it requires a larger power supply (typically 600 watts). If the server houses a large number of hard drives, it may require an even larger power supply.

Because the power supply is one of the most likely components to fail, many server computers have two built-in power supplies for redundancy. That way, if one of the power supplies fails, the other can pick up the load and keep the server running.

Considering Server Form Factors

Form factor refers to the size, shape, and packaging of a hardware device. Server computers typically come in one of three form factors: tower cases, rack-mounted servers, or blade servers. You can also get servers in very small packages; these are known as *tiny servers*.

Tower cases

Most servers are housed in a traditional tower case, similar to the tower cases used for desktop computers. A typical server tower case is 18 inches high, 20 inches deep, and 9 inches wide, with room inside for a motherboard, five or more hard drives, and other components.

Some server cases include advanced features specially designed for servers, such as redundant power supplies (so both servers can continue operating if one of the power supplies fails), hot-swappable fans, and hot-swappable disk-drive bays. (*Hot-swappable* components can be replaced without powering down the server.)

Rack-mounted servers

If you need only a few servers, tower cases are fine. You can just place the servers next to each other on a table or in a cabinet that's specially designed to hold servers. If you need more than a few servers, though, space can quickly become an issue. For example, what if your departmental network requires a bank of ten file servers? You'd need a pretty long table!

Rack-mounted servers are designed to save space when you need more than a few servers in a confined area. A rack-mounted server is housed in a small chassis that's designed to fit into a standard 19-inch equipment rack. The rack allows you to vertically stack servers to save space.

Blade servers

Blade servers are designed to save even more space than rack-mount servers. A *blade server* is a server on a single card that can be mounted alongside other blade servers in a blade chassis, which itself fits into a standard 19-inch equipment rack. A typical blade chassis holds six or more servers, depending on the manufacturer.

One of the key benefits of using blade servers is that you don't need a separate power supply for each server. Instead, the blade enclosure provides power for all its blade servers. Some blade server systems provide rack-mounted power supplies that can serve several blade enclosures mounted in a single rack.

In addition, the blade enclosure provides keyboard, video, and mouse (KVM) switching so that you don't have to use a separate KVM switch. You can control any of the servers in a blade server network from a single keyboard, monitor, and mouse. (For more information, see the sidebar, "Saving space with a KVM switch.")

SAVING SPACE WITH A KVM SWITCH

If you have more than two or three servers in one location, consider getting a KVM switch to save space by connecting several server computers to a single keyboard, monitor, and mouse. Then you can control any of the servers from a single keyboard, monitor, and mouse by turning a dial or by pressing a button on the KVM switch.

Simple KVM switches are mechanical and allow you to choose from 2 to 16 (or more) computers. More elaborate KVM switches can control more computers, using a pop-up menu or a special keyboard combination to switch among computers. Some advanced KVMs can even control a mix of PCs and Mac computers from a single keyboard, monitor, and mouse.

Another big benefit of using blade servers is that they drastically cut down the amount of cable clutter. With rack-mounted servers, each server requires its own power, keyboard, video, mouse, and network cables. With blade servers, a single set of cables can service all the servers in a blade enclosure.

Tiny servers

You can also get servers in very small packages — even as small as a deck of cards. Most of these computers are *single-board computers*, meaning that the entire computer is built on a single small motherboard that houses the CPU, RAM, disks, video, and network components. For Windows servers, NUC computers are an excellent choice; for Linux servers, look into Raspberry Pi.

Tiny servers are typically used for very specific purposes. For example, you might set up a NUC or Raspberry Pi to provide DHCP or routing services for a small network.

Understanding Virtualization

One final subject for this chapter is the concept of virtualization. Throughout this chapter, I use the term *server* to refer both to an operating system that provides services (such as file-sharing or directory services), as well as to the hardware on which that operating system runs. However, in many (if not most) modern network environments, a single physical computer system is used to run more than one *virtual machine* (VM). A VM is a simulation of an actual computer system. This concept is called *virtualization*. When virtualization is used, a single physical server computer actually runs more than one virtual server.

Virtualization is the reason that server computer hardware often has such high performance specifications, such as dual processors with multiple cores each and a large amount of RAM (256GB or more). In most environments, no single server really needs that much capacity. But when a single physical computer is responsible for running multiple virtual servers, the physical server must have sufficient capacity to run all its virtual servers.

Also, note that virtualization isn't just for servers: In many organizations, desktop computers are also virtualized. Virtualizing desktops offers many advantages, especially in organizations where most users need just basic computers to access a few simple business applications.

If this concept seems confusing at first, don't sweat it. You can find out more about virtualization in Book 3, Chapter 4, as well as in Book 5.